

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – I
PAPER –2
TEST DATE: 20-11-2022

ANSWERS, HINTS & SOLUTIONS

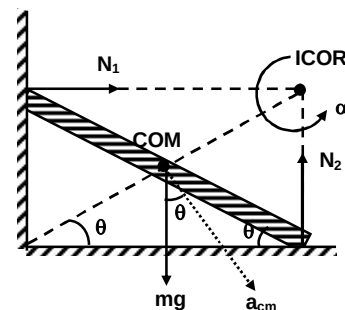
Physics

PART – I

Section – A

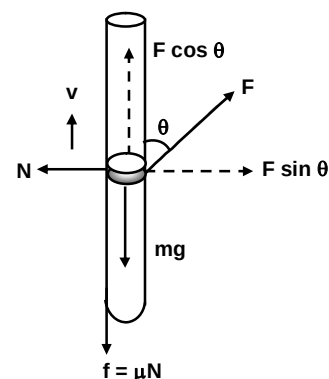
1.

Sol. D
 $\tau = I\alpha$ (about ICOR)
 $mg\left(\frac{\ell}{5}\right)\frac{1}{2} = \left(\frac{m\ell^2}{12} + m\left(\frac{\ell}{2}\right)^2\right)\alpha$
 $\alpha = \frac{6g}{5\ell}$
 $\therefore a_{cm} = \frac{\ell}{2}\alpha = \frac{3g}{5}$
 $mg - N_2 = (ma_{cm})\cos\theta$
 $mg - N_2 = \frac{3mg}{5}\left(\frac{4}{5}\right)$
 $\Rightarrow N_2 = \frac{13mg}{25}$



2.

Sol. A
 $\theta = \omega t$ ($\omega = \text{constant}$)
 $N = F \sin \omega t$
 $\frac{mdv}{dt} = (F \cos \omega t - \mu N - mg)$
 $\int_0^0 mdv = \int_0^T (F(\cos \omega t - \mu \sin \omega t) - mg) dt$
 $\frac{F}{\omega} [\sin \omega t + \mu \cos \omega t]_0^T = mgT$
 $\therefore \text{Also } \omega T = \frac{\pi}{2}$



$$\therefore F \left[\sin \frac{\pi}{2} + \mu \cos \frac{\pi}{2} - \mu \right] = mg \omega T$$

$$F[1 - \mu] = mg \frac{\pi}{2}$$

$$\therefore F = \frac{mg\pi}{2(1-\mu)}$$

3.

B

Sol.

F.B.D. of rod

$$\bar{\tau}_0 = 0 \quad (\text{about pivot})$$

$$\Rightarrow N_1 = \frac{mg}{2} \quad \dots(i)$$

F.B.D. of cylinder

$$\bar{\tau}_C = 0$$

$$\Rightarrow f_1 R = f_2 R$$

$$f_1 = f_2$$

 (say f)

$$\text{Also } \bar{\tau}_0 = 0$$

$$(N_1 + mg)\ell = N_2 \ell \Rightarrow N_2 = N_1 + mg$$

$$N_2 = \frac{3mg}{2} \quad \dots(iii)$$

$$\sum F_x = 0 \quad (\text{on the cylinder})$$

$$f(1 + \cos \theta) = N_2 \sin \theta$$

$$\therefore f = \frac{3mg \sin \theta}{2(1 + \cos \theta)}$$

$$\text{Also } f_{1\max} < f_{2\max} \quad (\because N_1 < N_2)$$

$$\therefore f \leq \mu N_1 \Rightarrow \frac{3mg \sin \theta}{2(1 + \cos \theta)} \leq \mu \frac{mg}{2}$$

$$\mu \geq \frac{3 \sin \theta}{1 + \cos \theta}$$

4.

A

Sol.

 F.B.D. of block is as shown ($\theta \leq 90^\circ$)

$$N = F \sin \theta + mg \cos \alpha$$

$$f = F \cos \theta - mg \sin \alpha$$

 for No sliding $f \leq \mu N$

$$F \cos \theta - mg \sin \alpha \leq \frac{1}{\sqrt{3}} (F \sin \theta + mg \cos \alpha)$$

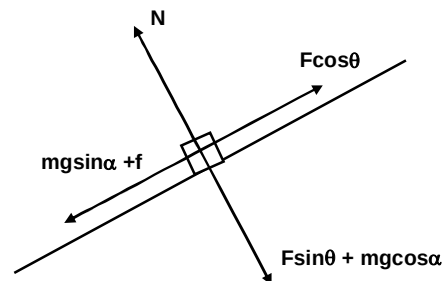
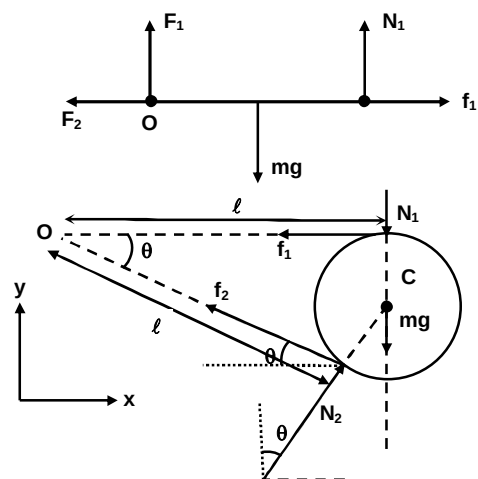
$$\cos \theta - \frac{\sin \theta}{\sqrt{3}} \leq \frac{mg}{F} \left[\sin \alpha + \frac{\cos \alpha}{\sqrt{3}} \right]$$

$$\cos \theta - \frac{\sin \theta}{\sqrt{3}} \leq 0 \quad (\because \text{no sliding even when } F \rightarrow \infty)$$

$$\tan \theta \geq \sqrt{3} \Rightarrow \theta \geq 60^\circ$$

 For $\theta > 90^\circ$

$$N = F \sin \theta + mg \cos \alpha$$



$$f = mg \sin \alpha - F \cos \theta$$

$$f \leq \mu N \Rightarrow \theta \leq 120^\circ$$

$$\therefore \text{Range} \Rightarrow \theta \in [60^\circ, 120^\circ]$$

5. A, B, D

Sol. Applying WET on bead from A to B $W_N + W_F = \Delta K$

$$O + F[(R-x) - (R+x)] = \frac{m}{2}[v^2 - u^2]$$

conservative

$$v^2 = u^2 - \frac{4Fx}{m} \Rightarrow u_0 = 2\sqrt{\frac{Fx}{m}}$$

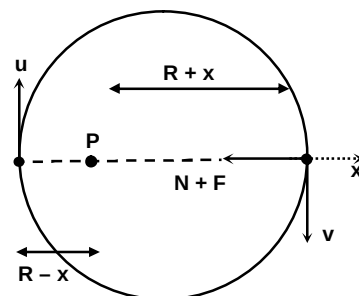
$$\text{Also } N + F = \frac{mv^2}{R}$$

$$N = \frac{m}{R}\left[u^2 - \frac{4Fx}{m}\right] - F = \frac{m}{R}[u^2 - u_0^2] - F$$

$$N = \frac{m}{R}[\alpha^2 - 1]u_0^2 - F = \frac{m}{R}[\alpha^2 - 1]\frac{4Fx}{m} - F$$

$$N = F[4\beta(\alpha^2 - 1) - 1]$$

$$\vec{N} = F[1 + 4\beta(1 - \alpha^2)]\hat{i}$$



6. A, D

Sol. As $PA + PB = 10\ell = \text{Constant}$

The bead will move along the elliptical path in vertical plane.

Such that $2a = 10\ell$ and $2ae = 8\ell \Rightarrow a = 5\ell, b = 3\ell$

$\therefore$ Equation of ellipse is

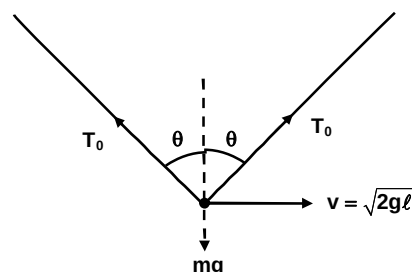
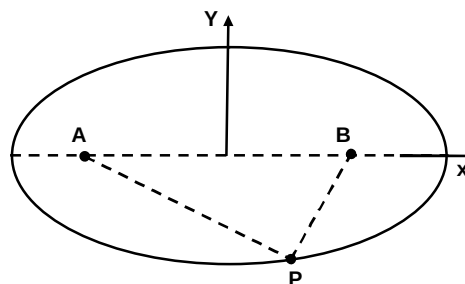
$$\frac{x^2}{(5\ell)^2} + \frac{y^2}{(3\ell)^2} = 1$$

$$r_0 = \frac{[1 + (y')^2]^{3/2}}{|y''|} \text{ at } (0, -3\ell) \quad (\because y' = 0, y'' = \frac{3}{25\ell})$$

$$\therefore r_0 = \frac{25\ell}{3}$$

$$2T_0 \cos \theta - mg = \frac{m}{r_0}(\sqrt{2g\ell})^2 \quad \left(\because \cos \theta = \frac{3}{5}\right)$$

$$T_0 = \frac{31mg}{30}$$



7. A, C

Sol. Angular momentum of system is conserved about the axis passing through the pivots

$$\vec{L}_i \text{ (Just before collision)} = \vec{L}_f \text{ (Just after collision)}$$

$$mv_0 R = \left(mR^2 + \frac{2}{5}(5m)R^2\right)\omega$$

$$\Rightarrow \omega = \frac{v_0}{3R}$$

Now, applying conservation of mechanical energy after the collision

$$\frac{1}{2} \left[mR^2 + \frac{2}{5} (5m)R^2 \right] \omega^2 + mgR = 5mg \frac{3R}{8}$$

$$\frac{mR^2}{2} (3) \frac{v_0^2}{9R^2} = \frac{7mgR}{8}$$

$$\text{Also } v_0 = \sqrt{2gh} \Rightarrow h = \frac{21R}{8} \Rightarrow \omega = \sqrt{\frac{7g}{12R}}$$

8. A, D

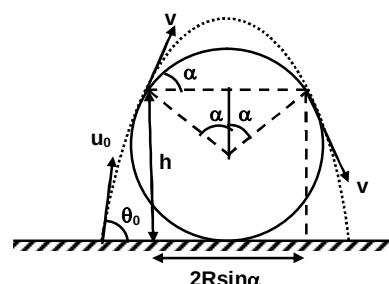
Sol. For u to be minimum, it just grazes the cylinder at two points as shown.

From COME

$$\frac{mu^2}{2} = \frac{mv^2}{2} + mgh$$

$$u^2 = v^2 + 2gh$$

$$u^2 = v^2 + 2gR(1 + \cos \alpha)$$



$$\text{Also } 2R \sin \alpha = v^2 \frac{\sin 2\alpha}{g} \Rightarrow v^2 = \frac{Rg}{\cos \alpha} \quad \dots(2)$$

From (1) and (2)

$$u^2 = Rg \left[2 + 2\cos \alpha + \frac{1}{\cos \alpha} \right]$$

$$\text{For } u \rightarrow \min. \quad \frac{d}{d\alpha} (u^2) = 0 \Rightarrow \cos \alpha = \frac{1}{\sqrt{2}} \Rightarrow \alpha = 45^\circ$$

$$u_0^2 = Rg [2 + 2\sqrt{2}]$$

$$\text{Also } u \cos \theta = v \cos \alpha$$

$$\tan^2 \theta_0 = [3 + 2\sqrt{2}]$$

9. A, D

Sol. Just after the jump by the frog.

According to COLM of (A + frog)

$$(4v_0 - v)m - mv = 0 \quad v = 2v_0$$

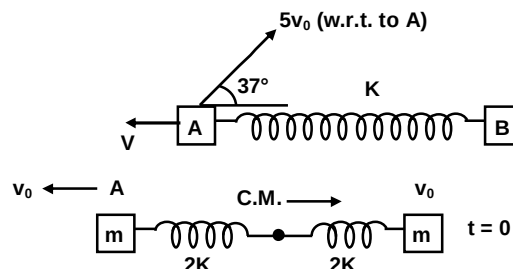
Now consider the motion of blocks in their COM frame.

At any time t

$$v_{BC} = v_B - v_{CM} = v_0 \cos(\omega t), \quad \omega = \sqrt{\frac{2k}{m}}$$

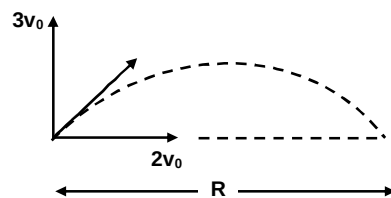
$$v_B = v_{CM} + v_0 \cos \omega t$$

$$v_B = -v_0 + v_0 \cos \omega t$$



Also the frog is in projectile motion from ground frame

$$R = 2 \left(\frac{3v_0}{g} \right) \frac{2v_0}{g} = \frac{12v_0^2}{g} \quad \left(T = \frac{6v_0}{g} \right)$$



Also frog land on B

$$\text{Then } \ell_0 + \int_0^T v_B dt = R$$

$$\ell_0 = R - \int_0^T v_0 (\cos \omega t - 1) dt$$

$$\ell_0 = R + v_0 T - \frac{v_0}{\omega} \sin \omega T$$

$$\therefore \ell_0 = \frac{12v_0^2}{g} + \frac{6v_0^2}{g} - v_0 \sqrt{\frac{m}{2K}} \sin \left(\sqrt{\frac{2K}{m}} \frac{6v_0}{g} \right)$$

$$\therefore \ell_0 = \frac{6v_0^2}{g} \left[3 - \frac{2}{\pi} \right].$$

10. B, C

$$\text{Sol. } F = mg + v_0 \frac{dm}{dt} \quad \text{also } m = \lambda y$$

$$F = \lambda yg + v_0 \lambda \frac{dy}{dt}$$

$$= \lambda yg + \lambda v_0^2$$

$$\text{Heat loss per unit time} = Fv_0 - \frac{dE}{dt}$$

$$= (\lambda yg + \lambda v_0^2) v_0 - \frac{d}{dt} \left[\frac{1}{2} (\lambda y) v_0^2 + \lambda yg \frac{y}{2} \right]$$

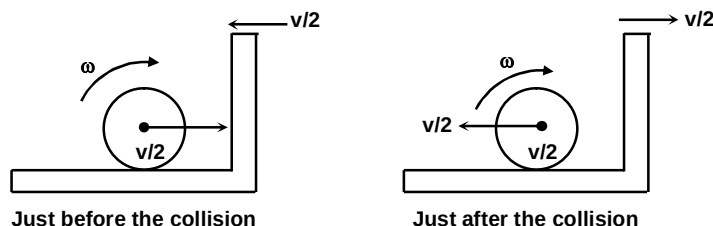
$$\lambda yg v_0 + \lambda v_0^3 - \frac{\lambda v_0^3}{2} - \lambda yg v_0$$

$$= \frac{\lambda v_0^3}{2}.$$

Section – B

11. 2

Sol. Observing the motion in COM frame. Then system will look like



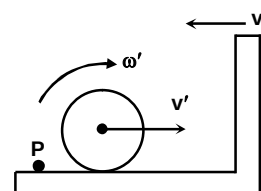
After the collision, ring starts sliding and then starts pure rolling on wedge as shown.

$$\omega'R = 2v' \text{ (pure rolling)}$$

Applying COAM of ring in centre of mass frame (after the collision) about point P on wedge

$$mR^2\omega + \left(-mR\frac{v}{2}\right) = mRv' + mR^2\omega'$$

$$mR\frac{v}{2} = 3mRv' \Rightarrow v' = \frac{v}{6}$$



Therefore the velocity of wedge is $\frac{v}{6}$ (in centre of mass frame) when pure rolling starts after the first collision. Then they again collide and ring again starts pure rolling on wedge and velocity of wedge is $\frac{v}{18}$ (in centre of mass frame) after 2nd collision.

The process goes on and there will be infinite collisions and finally wedge comes at rest in centre of mass frame.

$$\therefore \text{Velocity of wedge} = \frac{v}{2} \text{ (in ground frame)}$$

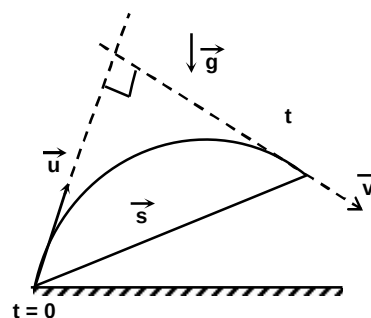
12. 5

Sol. $\vec{v} = \vec{u} + \vec{g}t$

$$\text{also } \vec{s} = (\vec{v}_{\text{ave}})t = \left(\frac{\vec{v} + \vec{u}}{2}\right)t$$

$$\therefore \vec{v} - \vec{u} = \vec{g}t \quad \dots(i)$$

$$\vec{v} + \vec{u} = \frac{2\vec{s}}{t} \quad \dots(ii)$$



Squaring and multiplying

$$(\vec{v} \cdot \vec{v} + \vec{u} \cdot \vec{u} + 2\vec{u} \cdot \vec{v})(\vec{v} \cdot \vec{v} + \vec{u} \cdot \vec{u} - 2\vec{u} \cdot \vec{v}) = 4g^2s^2$$

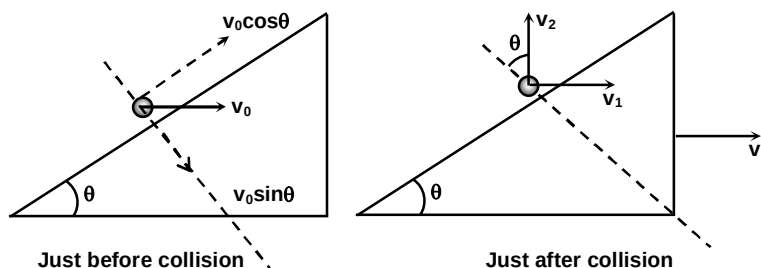
$$s^2 = \frac{(100 + 2\vec{u} \cdot \vec{v})(100 - 2\vec{u} \cdot \vec{v})}{4g^2}$$

s is max. when $\vec{u} \cdot \vec{v} = 0 \Rightarrow \vec{u} \perp \vec{v}$

$$\therefore s_{\text{max}} = \frac{100}{2(10)} = 5\text{m}$$

13. 6

Sol. If it strike again at same point, then the horizontal component of their velocities are same after the first collision.



As $e = 1$

$$v_{\text{sep}} = v_{\text{app}}$$

$$v_2 \cos \theta = v_0 \sin \theta$$

$$\therefore v_2 = v_0 \tan \theta = (40) \frac{3}{4} = 30 \text{ m/s}$$

$$t_0 = \frac{2v_2}{g} = 6 \text{ sec.}$$

14. 9

Sol.

$$T_{\max} = T_1$$

$$T_{\min} = T_2$$

$$T_1 - mg = \frac{mu^2}{l} \quad \dots(1)$$

$$T_2 + mg = \frac{mv^2}{l} \quad \dots(2)$$

From COME (of pendulum + earth system)

$$\frac{1}{2} mu^2 = \frac{1}{2} mv^2 + mg(2l)$$

$$\therefore u^2 = v^2 + 4gl \quad \dots(3)$$

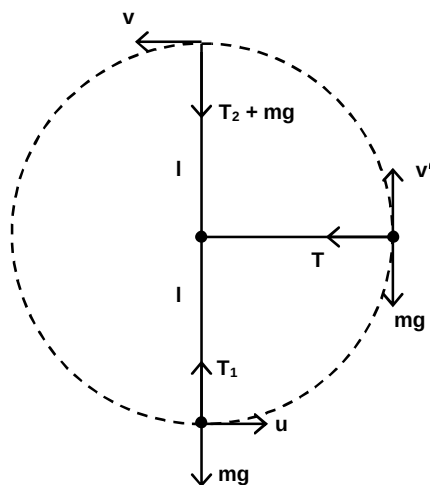
$$\text{also } \frac{T_1}{T_2} = 2 \quad \dots(4)$$

from (1), (2) (3) and (4)

$$u = \sqrt{11gl}, \quad v = \sqrt{7gl}$$

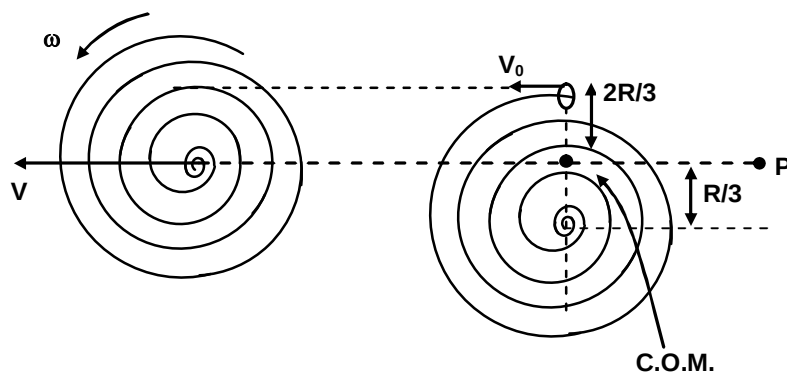
$$\therefore v' = \sqrt{9gl},$$

$$T = \frac{mv'^2}{l} = 9mg \quad K = 9$$



15. 2

Sol.



Applying COLM

$$mv_0 = (m + 2m)v \quad v = \frac{v_0}{3}$$

Applying COAM about point P

$$mv_0 \frac{2R}{3} = \left(\frac{MR^2}{2} \right) \omega \Rightarrow 2m \frac{v_0 R}{3} = mR^2 \omega$$

$$\therefore \omega R = \frac{2v_0}{3}$$

$$\therefore \frac{\omega R}{v} = 2$$

16. 3

$$\text{Sol. } T = \frac{2v_0 \cos\left(\frac{3\theta}{2}\right)}{g \cos\left(\frac{\theta}{2}\right)}$$

$$= \frac{2v_0}{g} \left[4 \cos^2\left(\frac{\theta}{2}\right) - 3 \right]$$

$$\text{Also } R \sin \theta = V_0 \sin 2\theta T$$

$$R \sin \theta = \frac{2v_0^2}{g} (2 \sin \theta \cos \theta) [2(1 + \cos \theta) - 3]$$

$$\text{As } v_0^2 = 2gh$$

$$\therefore h = \frac{R}{8[\cos \theta (2 \cos \theta - 1)]}$$

$$h \geq 0 \quad 2 \cos^2 \theta - \cos \theta \geq 0$$

$$\cos \theta (2 \cos \theta - 1) \geq 0$$

$$\cos \theta \geq \frac{1}{2}$$

$$\theta \leq \frac{\pi}{3} \quad K = 3.$$

2nd Method:

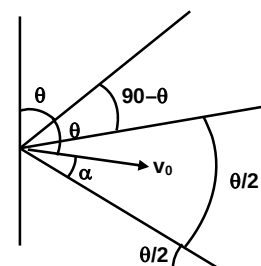
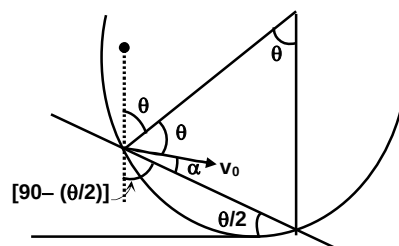
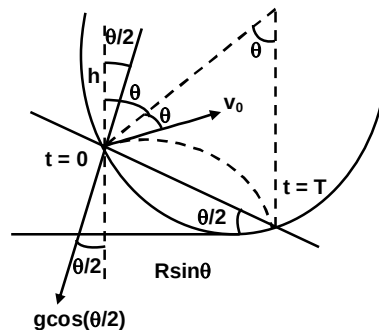
$$2\theta + \alpha + \left(90 - \frac{\theta}{2}\right) = 180^\circ$$

$$\alpha = 90 - \frac{3\theta}{2}$$

$$\text{Now } \alpha > 0$$

$$\theta < 60^\circ$$

$$\theta_{\max} = \frac{\pi}{3} \text{ radian}$$



17. 6

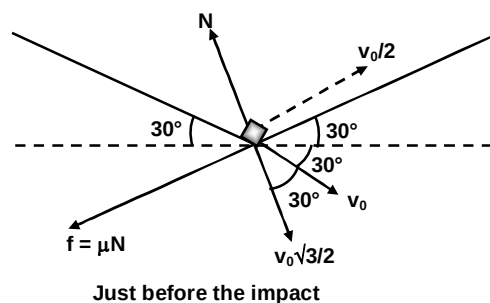
Sol. Applying impulse momentum theorem.

$$\int N dt = mv_0 \frac{\sqrt{3}}{2}$$

$$\int \mu N dt = m \left(\frac{v_0}{2} - v \right)$$

$$\therefore v = \frac{v_0}{2} (1 - \sqrt{3}\mu)$$

$$\text{Also } v_0^2 = 2gH \text{ and}$$

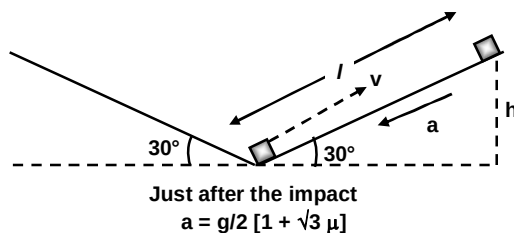


$$\therefore 2gh(1+\sqrt{3}\mu) = \frac{2gH}{4}(1-\sqrt{3}\mu)^2$$

$$\text{Also } \mu = \frac{1}{2\sqrt{3}}$$

$$h\left(1+\frac{1}{2}\right) = \frac{H}{4}\left(1-\frac{1}{2}\right)^2$$

$$\Rightarrow \frac{H}{4h} = 6$$



18.

8

Sol. In equilibrium, the centre of mass of the system must lie on the vertical line passing through hinge.

$$\therefore X_{CM} = 0$$

$$-m_1 \frac{L}{2} + m_2 \left(\frac{2R}{\pi} \right) = 0$$

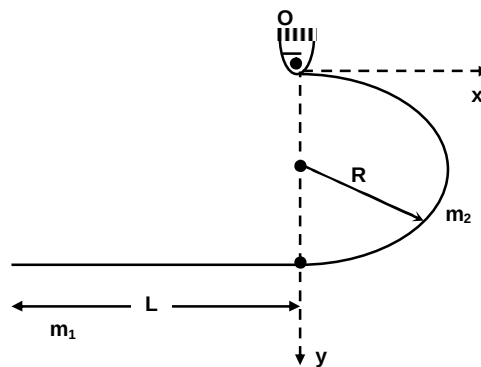
$$\frac{\lambda L^2}{2} = \lambda (\pi R) \frac{2R}{\pi}$$

$$\therefore L = 2R$$

$$Y_{CM} = \frac{m_1(2R) + m_2 R}{m_1 + m_2}$$

$$= \frac{\lambda L^2 + \lambda \pi R^2}{\lambda L + \lambda \pi R} = \left(\frac{\pi + 4}{\pi + 2} \right) \frac{L}{2} \Rightarrow \alpha = 4, \beta = 2$$

$$\therefore \alpha\beta = 8.$$

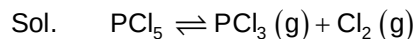


Chemistry

PART – II

Section – A

19. A



$$\alpha = \frac{D-d}{(n-1)d}, \quad n = 2$$

$$(1+\alpha) = \left(\frac{D}{d}\right)$$

 Thus, $(1+\alpha)$ will linearly increases with increase of $\left(\frac{D}{d}\right)$.

20. A

Sol. $\ell n K = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{R}$

$$\Delta S^\circ = 8R$$

21. B

Sol. Change in potential energy $= -\frac{2x}{8} - (-2x) = -\frac{2x}{8} + 2x = \frac{-2x + 16x}{8}$
 $= +\frac{14x}{8} = +\frac{7x}{4}$

22. A

Sol. $K_{298} = \frac{2.303}{t_{1/2}} \log \frac{100}{50}$
 $K_{308} = \frac{2.303}{t_{1/2}} \log \frac{100}{25}$
 $\frac{K_{298}}{K_{308}} = \frac{\log 2}{\log 4} = \frac{1}{2} = 0.5$

23. A, B

 Sol. Possible transition $3 \rightarrow 2$ and $4 \rightarrow 2$

$$\lambda_{3 \rightarrow 2} \Rightarrow \frac{36}{5R}$$

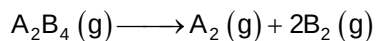
$$\lambda_{4 \rightarrow 2} \Rightarrow \frac{16}{3R}$$

24. A, B, C

 Sol. Molecular mass of $\text{A}_2\text{B}_4 = 100$

$$2A + 4B = 100, \quad \frac{2A + 4B}{2A} = 5 \text{ (given)}$$

$$A = 10, B = 20, \quad \frac{A}{B} = \frac{10}{20} = 1:2$$



$$1 - \alpha \qquad \qquad \alpha \qquad \qquad 2\alpha$$

At 10 min. V.D. = 25

Molecular mass = 50

$$50 = \frac{100}{1 + 2\alpha}$$

$$\alpha = \frac{1}{2} \quad \%B_2 \text{ (by mole)} = \frac{1}{2} \times 100 = 50\%$$

$$\%A_2 \text{ (by mole)} = \frac{1}{4} \times 100 = 25\%$$

At t = 20 min.

$$40 = \frac{100}{1 + 2\alpha}$$

$$\alpha = 0.75$$

$$-\frac{d(A_2B_4)}{dt} = \frac{0.5 - 0.25}{10} = \frac{0.25}{10}$$

$$+\frac{d(A_2)}{dt} = \frac{0.75 - 0.50}{10} = \frac{0.25}{10}$$

Ratio of rate = 1 : 1

25. A, C, D

Sol.



$$\text{At equilibrium} \quad 4 - 2x \quad x - 3y \quad x$$



$$\text{At equilibrium} \quad 4.25 - 2y \quad x - 3y \quad 4y$$

Given x = 1.0 M

$$4y = 0.5 \text{ M}$$

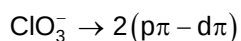
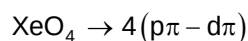
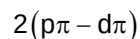
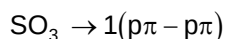
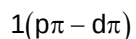
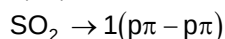
$$\text{Using these value } K_{C_1} = \frac{5}{32}$$

$$K_{C_2} = \frac{4^5}{5^6}$$

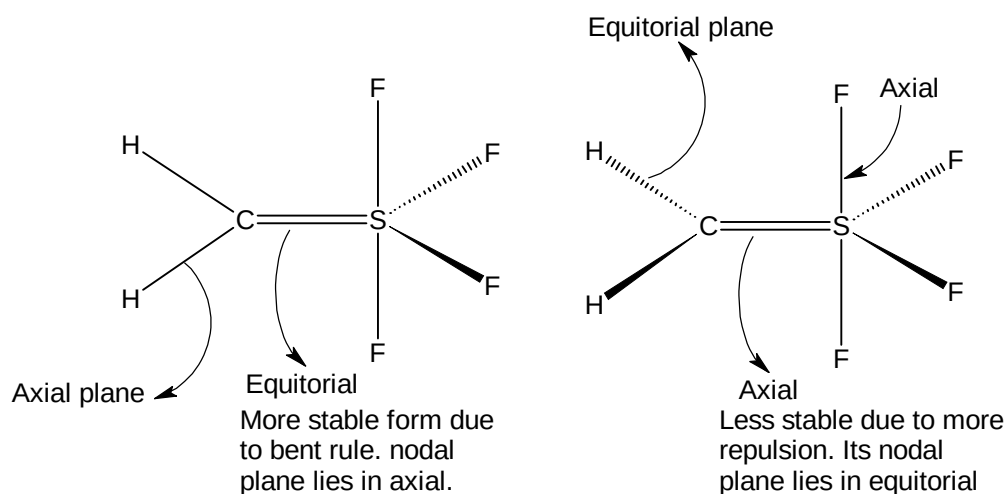
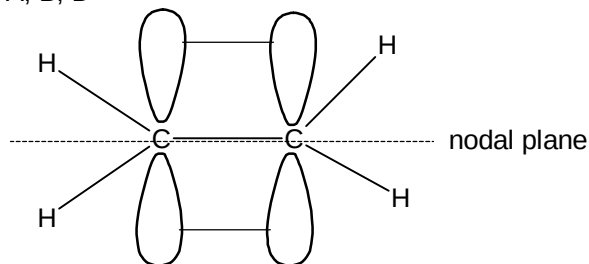
$$\sqrt{K_{C_2}} = \left(\frac{32}{125} \right)$$

26. B, C, D

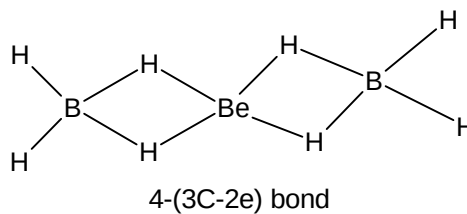
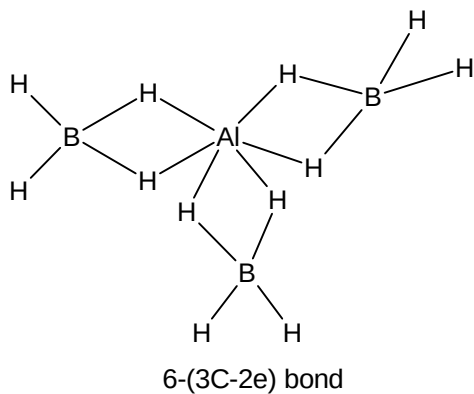
Sol.



27. A, B, D
Sol.



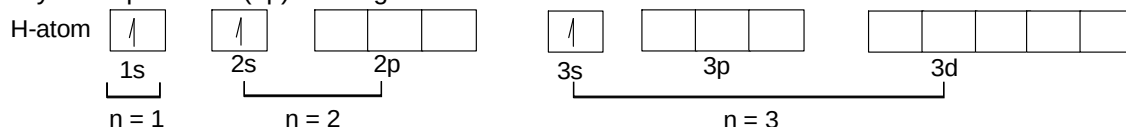
28. C, D
Sol. $(\text{AlCl}_3)_n \rightarrow$ has $(3C - 4e)$ bond not $(3C - 2e)$
 $(\text{B}_2\text{H}_6) \rightarrow$ Two $(3C - 2e)$ bond

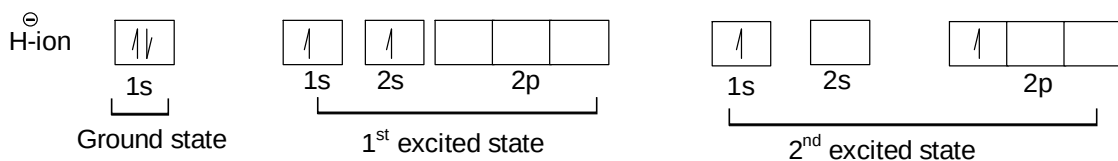


Section - B

29. 5
Sol. $x = 9, y = 3, z = 2$

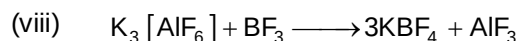
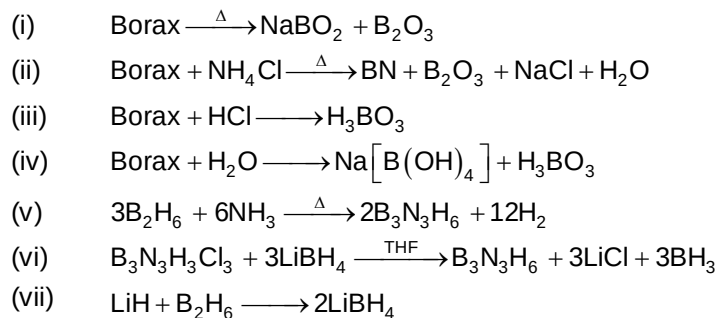
In one electron system all orbitals of a shell are degenerate (same energy level). In case of many electron system different orbitals of a shell are non-degenerate. Hence, in second excited state only three p-orbitals (2p) are degenerate.





30. 6

Sol.



Except (vii) and (viii) all options can give sp^2 hybridised Boron centre as one of the product.

31. 6

Sol.

Lone pair in $[\text{TeBr}_6]^{2-}$, $[\text{IF}_2]^+$, SNF_3 and $[\text{XeF}_3]^-$ are 1, 2, 0 and 3 respectively. Hence, $x = 6$.

Diamagnetic species are Li_2 , C_2 , O_2^{2-} , K_2O , N_2O_4 , $\text{B}_3\text{N}_3\text{H}_6$.

Note: Be_2 does not exist so we can't say it is paramagnetic or diamagnetic species.

Hence, $y = 6$.

$$\frac{x+y}{2} = 6.$$

32. 4

Sol.

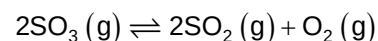
In container (I)

In container (II)

KMnO_4 will oxidized only SO_2
 Moles of O.A. = Moles of R.A.

$$\frac{0.2 \times 4}{2} = \frac{\text{Moles of } \text{SO}_2}{5}$$

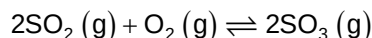
$$\text{Mole of } [\text{SO}_2]_{\text{equilibrium}} = 2$$



$$10 - 2x_1 \quad 2x_1 \quad x_1$$

$$2x_1 = 2$$

$$x_1 = 1.0$$



$$2x_1 \quad x_1 \quad 0$$

$$2x_1 - 2a \quad x_1 - a \quad 2a$$

$$\text{Total moles } n_T = (3x_1 - a)$$

$$\text{Given } \frac{x_1 - a}{3x_1 - a} = \frac{1}{4} \quad a = \frac{x_1}{3}$$

$$y = K_P = \frac{\left(\frac{2a}{n_T} \times 20\right)^2}{\left(\frac{2x_1 - 2a}{n_T} \times 20\right)^2 \left(\frac{x_1 - a}{n_T} \times 20\right)} = \left(\frac{1}{20}\right) \text{ atm}^{-1}$$

$$x = K_c \frac{\left(\frac{2}{5}\right)^2 \left(\frac{1}{5}\right)}{\left(\frac{8}{5}\right)^2} = \frac{1}{80} (\text{mol} / \ell) \quad \text{Value of } \frac{y}{x} = \frac{80}{20} = 4$$

33. 6

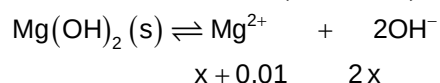
Sol. $a = 4 \rightarrow \text{i, iv, vi, vii}$
 $b = 2 \rightarrow \text{ii, ix}$
 $v \text{ and viii forms } \delta \text{ M.O.}$
 $d = 2 \rightarrow \text{iii, x}$
 $\frac{a^2 + b^2 + 2d}{4} = \frac{24}{4} = 6$

34. 9

Sol. I, II, III and IV are correct due to bent rule and V is incorrect. Correct order in case of V is $\text{II} > \text{III} > \text{IV} > \text{I} = \text{V}$.
 $x = 4$
 $y = 1$
 $2x + y = 9$

35. 8

Sol. K_{sp} of $\text{Mg}(\text{OH})_2$ is $= \left(\frac{5.8}{58} \times 10^{-3}\right)^3 \cdot 4 = 4 \times 10^{-12}$



$$(x + 0.01)(4x^2) = 4 \times 10^{-12}$$

$$4x^2 \times 0.01 = 4 \times 10^{-12}$$

$$x = 10^{-5} \text{ mole} / \ell$$

$$[\text{Mg}^{+2}] \text{ in container II} = 1.0 \times 10^{-5}$$

$$\text{For basic buffer } \text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_4\text{OH}]}$$

$$\text{pOH} = 5 + \log 5$$

$$= 5.70$$

$$\text{OH}^- = 2 \times 10^{-6}$$

$$[\text{Mg}^{+2}] = \frac{4 \times 10^{-12}}{4 \times 10^{-12}} = 1.0 \text{ M}$$

$$[\text{Al}^{3+}] = \frac{1.6 \times 10^{-20}}{8 \times 10^{-18}} = 2 \times 10^{-3} \text{ M}$$

$$x = 10^{-5}$$

$$y = 1.0$$

$$z = 2 \times 10^{-3}$$

$$\text{Value of } \frac{yz}{25x} = \frac{200}{25} = 8.$$

36. 5

Sol. Statement 4, 11 and 12 are incorrect and other statements are correct.
 $x = 9, y = 3$.

Mathematics**PART – III****Section – A**

37. A

Sol. By differentiating, $f'(x) + \frac{1}{1+x^2}f(x) + \frac{\tan^{-1}x}{1+x^2} = 0$

$$f(x) = 1 - \tan^{-1}x - 2e^{-\tan^{-1}x}$$

38. A

Sol. $\int_1^a (f(x) + |f(x)|) dx = g(a)$; $f(x) = 2x^3 + 6x - 8$

$$\int_2^3 \left| \frac{f(x)}{g(x)+5} \right| dx = \frac{1}{2} \ln \left(\frac{101}{22} \right)$$

39. D

Sol. $\int e^{x^2} \left(\frac{2x - \cot x}{\sin x} \right) \cdot \ln \left(\frac{e^{x^2}}{\sin x} \right) dx$

$$\text{Let } \frac{e^{x^2}}{\sin x} = t \Rightarrow \int \ln(t) dt = t(\ln t - 1) + c$$

40. B

Sol. $x^2 + y^2 - ax = 0 \Rightarrow x^2 + 2xy \cdot y' - y^2 = 0$

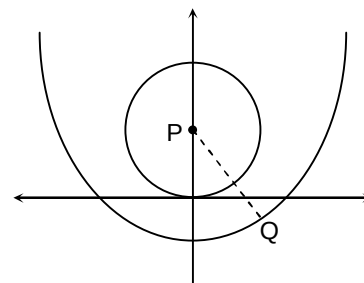
$$\text{Orthogonal curve : } x^2 - \frac{2xy}{y'} - y^2 = 0 ; y' = \frac{2xy}{x^2 - y^2}$$

$$\text{By solving, } x^2 + y^2 = \frac{y}{c}$$

$$\text{Passing through } (1, 1) \text{ so } x^2 + y^2 = 2y$$

$$\text{For shortest distance, } Q = (1, 0)$$

$$D = PQ - r = \sqrt{2} - 1$$



41. A, B, D

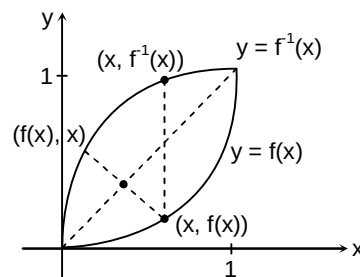
Sol. $f''(x) = f(x) \cdot (\tan x - 1)(\tan^3 x - \tan^2 x + 3 \tan x + 1)$

$$g(x) = \tan^3 x - \tan^2 x + 3 \tan x + 1 = 0 \text{ has exactly one root in } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \text{ which lies in}$$

$$x \in \left(-\frac{\pi}{4}, 0 \right)$$

$$\text{So, } f''(x) = 0 \text{ at } x = \frac{\pi}{4} \text{ and } k \text{ where } k \in \left(-\frac{\pi}{4}, 0 \right)$$

42. A, C

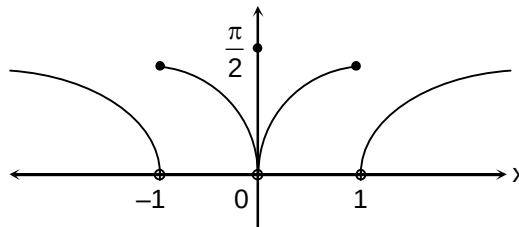
Sol. $f(x) \cdot f^{-1}(x) < x$ 

43. A, B

Sol. $f'g' = f'g + fg'$ by solving $g(x) = e^{-\frac{1}{x-2}}$
 $g(x)$ is discontinuous at one point at $x = 2$

44. B, D

Sol. Graph of $y = f(x)$



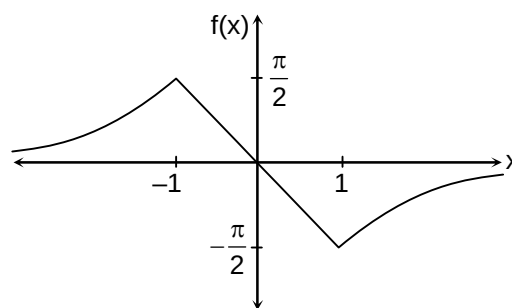
45. A, D

Sol. $f(x) = \int_0^{3\pi/4} \left| x - \frac{3 \sin t}{\cos t + \sqrt{2}} \right| \left(\frac{1 + \sqrt{2} \cos t}{(\cos t + \sqrt{2})^2} \right) dt$

$$f(x) = \int_0^1 |x - 3u| du \Rightarrow f(x) = \begin{cases} \frac{3}{2} - x & ; x \leq 0 \\ \frac{x^2}{3} - x + \frac{3}{2} & ; 0 \leq x \leq 3 \\ x - \frac{3}{2} & ; x \geq 3 \end{cases}$$

46. B, C

Sol. $\frac{x^2 - x + 1}{x^2 + x + 1} \in \left[\frac{1}{3}, 3 \right] \forall x \in \mathbb{R}$
 Maxima at $x = -1$ and minima at $x = 1$



Section - B

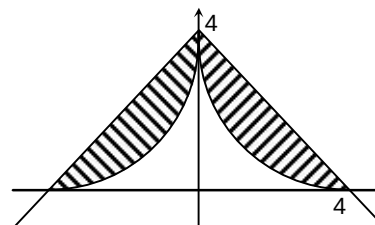
47. 1

Sol. $\int \frac{(6xf(x) - 4x^2) - 2f(x)}{(3f(x) - 2x)(x^2 - f(x))^2} dx = \int \frac{2x}{(x^2 - f(x))^2} dx - \int \frac{f'(x)}{(x^2 - f(x))^2} dx = -\frac{1}{x^2 - f(x)} + c$

Substitute $f(x) = \frac{2}{4x - 3f(x)}$

48. 2

Sol. $\frac{16\lambda}{3} = 2 \left[8 - \int_0^4 (x - 4\sqrt{x} + 4) dx \right]$



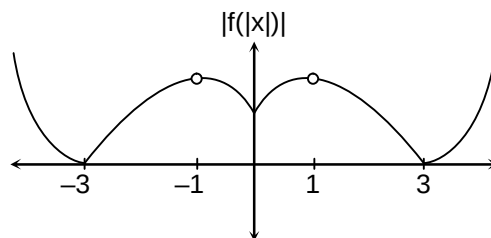
49. 5

Sol.
$$\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \left(\frac{n^2}{n^2 + r^2} \left(\lim_{m \rightarrow \infty} \sum_{k=1}^m \left(\frac{1}{n^2 + \frac{k^2}{m^2}} \right) \frac{1}{m} \right) \right) \right) = \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{n^2}{n^2 + r^2} \int_0^1 \frac{1}{n^2 + x^2} dx \right)$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{n}{n^2 + r^2} \tan^{-1} \left(\frac{r}{n} \right) \right) = \int_0^1 \frac{\tan^{-1} x}{1 + x^2} dx = \frac{\pi^2}{32}$$

50. 7

Sol. $f(x) = \frac{(x^2 - 1)(x^2 - 9)}{(x - 1)(x + 3)} \quad x \in \mathbb{R} - \{-3, 1\}$

Graph of $y = |f(x)|$ Discontinuity at $x = \pm 1$ Non-differentiable at $x = 0, \pm 1, \pm 3$ 

51. 8

Sol. $I = \int_0^{4\pi} \frac{dx}{2 + \sin 4x} = \int_0^{4\pi} \frac{dx}{2 - \sin 4x} \Rightarrow I = 2 \int_0^{4\pi} \frac{1}{4 - \sin^2 4x} dx = 64 \int_0^{\pi/8} \frac{1}{4 - \sin^2 4x} dx$

$$I = 64 \int_0^{\pi/8} \frac{2}{7 + \cos 8x} dx ; 3x = t \Rightarrow I = \frac{128}{3} \int_0^{3\pi/8} \frac{1}{7 + \cos \left(\frac{8t}{3} \right)} dt$$

52. 4

Sol. $f(x) = k(x + 1)^2(x - 1)^2(x - 5)^2 + 4 \Rightarrow f(2) = 81k + 4 = 85 \Rightarrow k = 1$

$$f(x) = (x + 1)^2(x - 1)^2(x - 5)^2 + 4 \Rightarrow f'(x) = 2(x + 1)(x - 1)(x - 5)(3x^2 - 10x - 1)$$

$$\alpha + \beta = \frac{10}{3} \Rightarrow \lim_{x \rightarrow \frac{10}{3}} \left(\frac{x^2 - 9}{x^2 - 5x + 6} \right) = \frac{19}{4}$$

53. 4

Sol.
$$\lim_{x \rightarrow 0} \frac{\frac{2022 \left((1 + x^{2022})^{\frac{1}{2021}} - 1 \right)}{x} - \frac{2023 \left((1 + x^{2023})^{\frac{1}{2021}} - 1 \right)}{x}}{2022x^{2021}} = \frac{1}{2021}$$

$$\lambda = 2021 = 43 \times 47 \text{ number of divisors} = 2 \times 2 = 4$$

54. 2

Sol. Area = $2 \int \left(\frac{y^2 + 3}{4} - \sqrt{y} \right) dy = \frac{1}{3}$

